

Ductwork System Spec Sheet

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Project: Lakefront Office Tower — Phase II

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SECTION 23 31 13 — METAL DUCTS

1.0 SCOPE OF WORK

Furnish and install all sheet metal ductwork, fittings, VAV terminal units, diffusers, registers, and grilles per drawings M2.11–M2.18. All ductwork constructed and installed per SMACNA HVAC Duct Construction Standards 3rd Ed.

2.0 EQUIPMENT SCHEDULE

Duct System	Pressure Class	Material	Gauge	Joints	External Insulation
Supply Main	2-in. SP	G90 Galv. Steel	20 ga.	TDC/TDF	2-in. duct wrap R-8
Supply Branch	2-in. SP	G90 Galv. Steel	22 ga.	TDC/TDF	1.5-in. duct wrap R-6
Return Main	1-in. SP	G90 Galv. Steel	22 ga.	S&D	None (plenum)
Return Branch	1-in. SP	G90 Galv. Steel	24 ga.	S&D	None
Exhaust	1-in. SP	G90 Galv. Steel	24 ga.	S&D	None
Kitchen Exhaust	2-in. SP	304 SS	16 ga.	Welded	1-hr fire-wrap
OA / Relief	1-in. SP	G90 Galv. Steel	22 ga.	TDC/TDF	2-in. R-8 (exterior)

3.0 SYSTEM SCHEMATIC / DRAWING

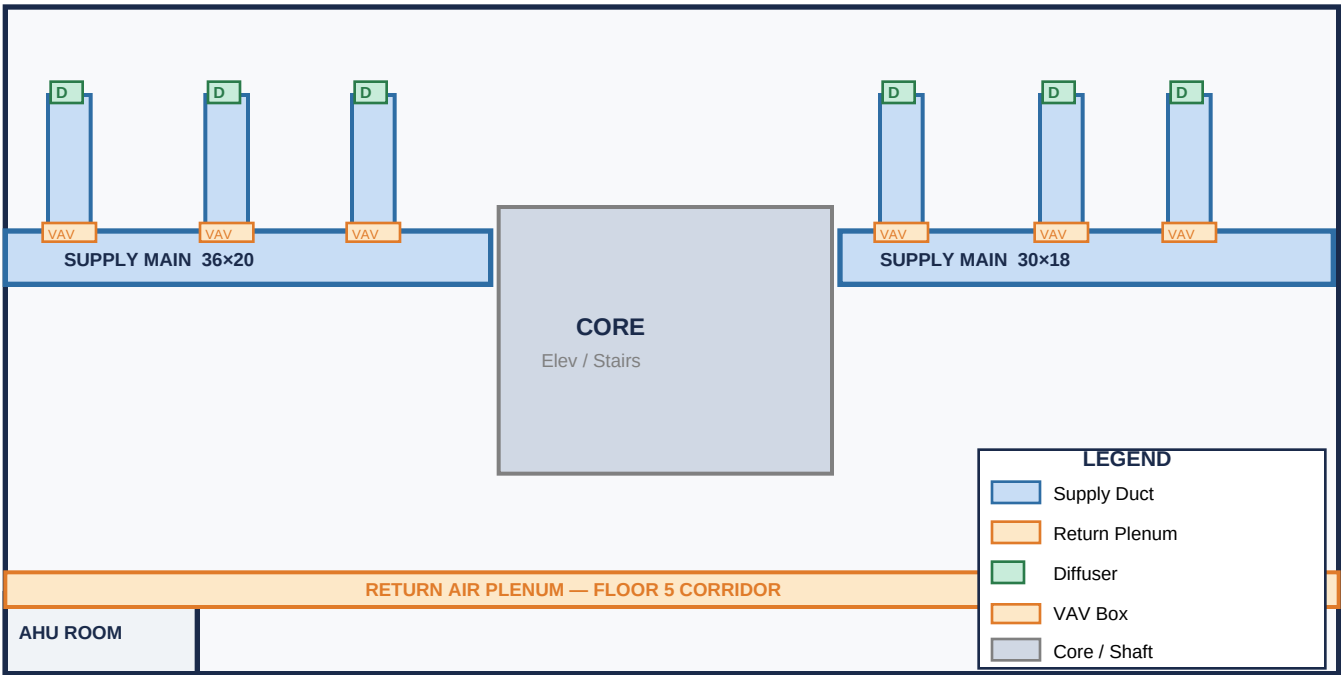


Figure 2 — Typical Floor Duct Layout, Floor 5 (Schematic — Not to Scale)

4.0 TECHNICAL SPECIFICATIONS

4.1 VAV Terminal Units

Type	Single-duct pressure-independent DDC VAV box
Construction	Galvanized steel, 20-ga minimum
Controls	24V electronic actuator, BACnet MS/TP
Reheat	Hot water coil, 2-row copper/Al fins where scheduled
Range	200–2,400 CFM (see VAV Box Schedule, Dwg M2.15)

4.2 Diffusers & Registers

Supply Diffusers	4-way adjustable bar, aluminum, white powder coat — size per Dwg M2.16
Return Grilles	Fixed bar, aluminum, white powder coat — face vel. $\leq$ 500 FPM
Exhaust Registers	Fixed bar, aluminum — face vel. $\leq$ 400 FPM
Balancing	All devices provided with integral volume adjustment dampers

5.0 INSTALLATION NOTES

- All duct joints sealed with UL 181A/B mastic or foil-faced tape rated for pressure class.
- Flexible connections: max 18-in. length; no sags, bends, or restrictions.
- Rectangular duct >19-in. width: intermediate hangers at 4-ft max spacing.
- Access doors min 12×12 in. at all fire/smoke dampers, coil valves, and IAQ sensors.
- Leakage test: pressure class 2 systems per SMACNA Leakage Class 6 (max 6 cfm/100 sf).
- Vapor barrier facing must be continuous, lap-sealed, and un-punctured.